Functional Specification

Pages, actions, rules and acceptance criteria

Version: 1.0 · **Status:** Implementation handoff · **Date:** 22 October 2026

1. Shared application shell

- Global sidebar navigation filtered by permission.
- Top bar with page title/breadcrumb, notification access and user menu.
- Search/filter patterns consistent across case lists.
- Status labels, due-date indicators and empty/error/loading states.
- All data fetched from authorised backend services.

2. Public landing page

Purpose: explain the prototype and direct users to sign in.

- Product name and concise purpose.
- Capability summaries: approval workflow, document pre-check, compliance tracking, alerts/audit.
- Sign-in action.
- Footer disclaimer that this is a prototype and not an official AICTE service unless authorised.

Acceptance: no fake official seal, endorsement or live integration claim.

3. Login and access

- Sign in with configured authentication.
- Load role and scope from the server-authenticated session.
- Route to institution, officer or administrator workspace.
- Provide sign-out and expired-session handling.
- Deny access if the role is inactive or scope is missing.

Acceptance: changing a URL or client-side role value cannot grant additional permissions.

4. Institution dashboard

- Summary counts for drafts, submitted, returned, awaiting action and completed.
- Lists of applications, next action, due date and last update.
- Upcoming compliance obligations and document renewals.
- Notifications and links to unresolved deficiencies.
- Create/open application action.

Acceptance: institution only sees its own records and explicitly authorised related records.

5. Application creation and tracking

- Select an enabled application type.

- Show a checklist based on a versioned configured rule set.
- Save draft, upload supporting documents and submit.
- Show application reference, current stage, stage history and next action.
- Display the route and required steps only when the configured route supports disclosure.

Acceptance: submission checks required fields/files; it does not bypass formal validation or official review.

6. AI pre-submission document checker

- Upload supported file type/size.
- Show file validation and text extraction state.
- Run configured field/date/completeness/legibility checks.
- Show each finding with severity, rule/source, evidence/page where available, explanation and suggested correction.
- Allow replacement upload and rerun.
- Clearly distinguish `Potential issue`, `Not checked`, `Extraction failed`, `Manual review needed` and `Passed checks`.
- Preserve prior file versions and check runs.

Acceptance: no “guaranteed approval” message; no claim of authenticity without authorised verification.

7. Deficiency and correction centre

- Officer records deficiency reason, applicable requirement/source, requested evidence, responsible party and due date.
- Institution views deficiency and submits correction.
- Officer reviews correction and records resolved, still deficient or needs clarification.
- Keep each attempt and response in the case timeline.

Acceptance: unresolved deficiency remains open; a file upload alone does not mark it resolved.

8. Officer dashboard and scrutiny workspace

- Assigned queue with filters by stage, age, due date and status.
- Open case with application details, document versions, pre-check findings and audit history.
- Record observations and return/revert with required reason where applicable.
- Execute permitted workflow transition.
- Review corrected evidence.

Acceptance: only authorised roles can take each action; every decision/action records actor and time.

9. EVC management

- Identify applications configured as requiring EVC.
- Place them in the EVC queue only when prerequisites are complete.
- Record readiness, proposed/confirmed schedule, completion and authorised outcome.
- Alert on approaching or exceeded configured timeline.
- Escalate to configured recipient.

Acceptance: a case cannot be marked EVC-complete without required authorised completion data.

10. Undertaking, compliance and document validity

- Create an obligation tied to institution and, where applicable, approval/case.
- Record condition, evidence needed, due date, source/rule and responsible party.
- Record lease/other prescribed document validity and expiry when applicable.
- Generate advance reminders according to configured policy.
- Accept renewal/evidence submission and route for authorised verification.
- Track `Due`, `Submitted`, `Awaiting verification`, `Verified`, `Rejected/needs correction`, `Overdue`, `Closed`.

Acceptance: submission and verification are separate states; expired documents are not automatically treated as legally invalid beyond the configured rule's meaning.

11. Next-action, reminders and escalation

- Every pending case/task shows next actor, action and due date when defined.
- Scheduled backend job identifies upcoming/overdue actions.
- Send in-app notifications and optional configured email notifications.
- Escalate according to a versioned policy.
- Record sent/failed/retried/acknowledged state.

Acceptance: repeated job execution does not create duplicate alerts; no unconfigured statutory deadline is invented.

12. Administrator dashboard

- Aggregated counts by permitted region, stage and status.
- Ageing and overdue lists.
- Workload by authorised team/officer.
- EVC status, compliance obligations, upcoming/expired documents and escalations.
- Drill-down to permitted cases.
- Export/report actions subject to permission and data-minimisation rules.

Acceptance: aggregates and drill-down rows respect region and role scope.

13. Configuration and audit

- Manage versioned route definitions, deadlines, notification rules and escalation policies where authorised.
- Each rule records source reference, effective dates, version, authoriser and change summary.
- View audit trail for transitions, observations, submissions, verification, escalations and configuration changes.
- Restrict user/role administration to authorised administrator permissions.

Acceptance: a configuration change cannot silently rewrite historic audit events or past due dates.

14. Reports

- Application pendency and ageing by stage/region.
- EVC queue, scheduled, completed and delayed cases.
- Undertaking/compliance status.
- Documents approaching expiry, expired and awaiting renewal verification.
- Overdue and escalation outcomes.

- Workload/performance summaries.

Acceptance: every report displays the selected scope, date range and data freshness; exports contain only authorised data.

15. Global error and empty states

- No results: explain how to change filters.
- No permission: explain access is restricted without exposing protected details.
- Failed upload: explain supported file types/size and offer retry.
- AI/OCR failure: preserve uploaded file and offer retry/manual review.
- Network/server failure: show safe error and retry option.
- Stale/concurrent edit: prompt refresh and avoid silent overwrite.

16. Minimum end-to-end demo acceptance script

1. Institution demo user signs in and creates a sample application.
2. Upload a synthetic sample document and run pre-check; show at least one explainable warning.
3. Submit the application; officer sees it in the correct queue.
4. Officer returns it with a reason and due date.
5. Institution uploads a correction; the previous version remains in history.
6. Officer reviews the correction and routes the case to the next configured stage.
7. Demonstrate an EVC-required case entering the queue and a deadline reminder.
8. Demonstrate an undertaking or lease document approaching expiry, evidence submission and pending verification.
9. Trigger an overdue escalation and show the event in the audit trail.
10. Administrator sees authorised aggregate metrics and can inspect the case history.
11. Test unauthorised access and invalid stage transitions; both must be rejected.
12. Demonstrate that all test data and simulated integrations are labelled as demo data.